

xMCF

ISO/PAS 8329

Description of mechanical connections
and joints in structural systems

ISO/PAS 8329 xMCF WG meeting
focus group on issues [#105](#) / [#61](#)
2026-01-21

Agenda

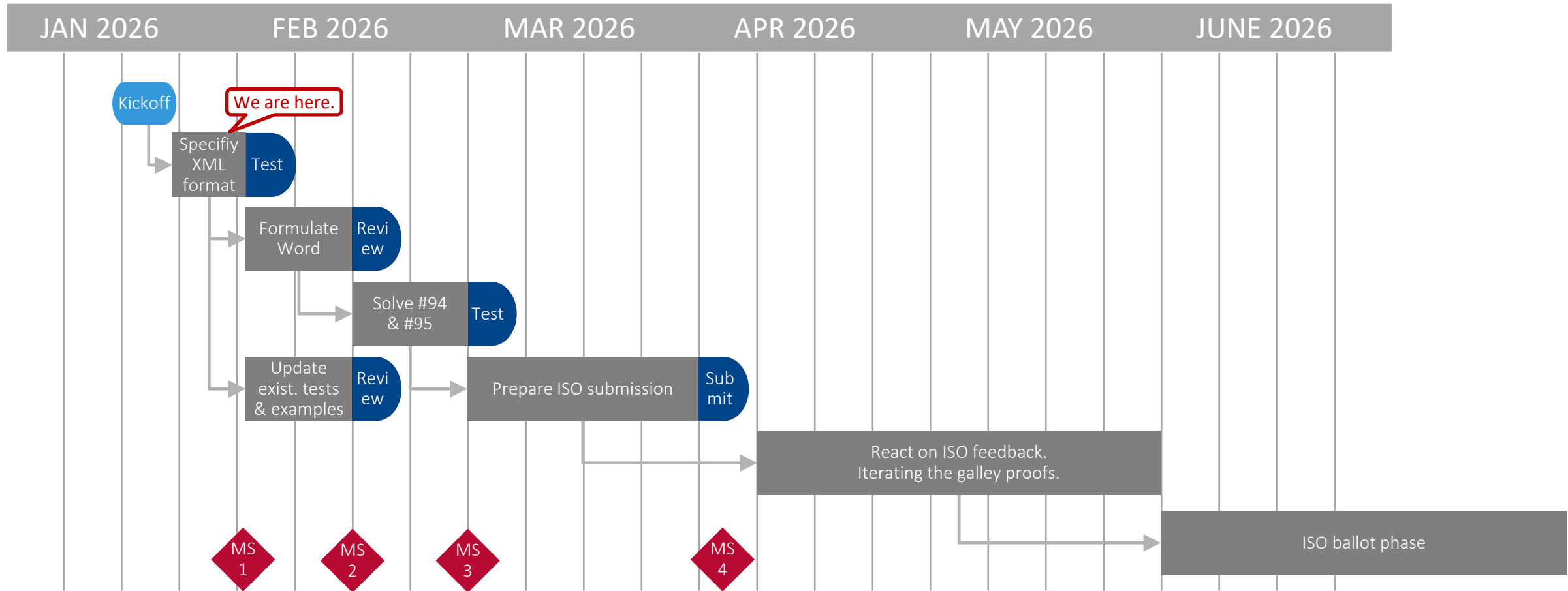
As announced at https://github.com/economidis-nick/createXSDforxMCF/blob/master/VDA_FAT_AK_25/Meetings/2026-01-21_VideoConference

Today's objective: **Create XML format suggestions**

- Welcome
- Timeline for next release
- Focus on detailing the XML elements, based e.g. on the [comment in GitHub](#) at issue [#105](#).
- Steps we should take next



Timeline for a release in the summer



Updating the PAS (or else striving for an IS, directly) shall be cross-checked with the chair of ISO/TC 184 / SC 4.
PPC was postponed to 18th of Feb.

Detailing XML Elements — Catalog Syntax, Screw Example

In the last meeting, we identified the example XML file [chapter 9 5 7 exampleA patched for xMCF3.2.xml](#) (provided by the comment of [2026-10-14 09:17+0100](#)) as a good starting point.

It suggests as a syntax for defining an *internal catalog* at XML root level:

```
<internal_catalog>
  <connection_0d label="screw_catalog_item_4711">
    <threaded_connection length="50." diameter="10"
      head_diameter="16.0" head_height="5" sink_size="4" thread_length="35"
      torque="80" angle="30" pretension="1200" part_code="M10x50 12.9" >
      <screw /> <!-- Screw may come without any attributes -->
      <washer outer_diameter="20"/>
    </threaded_connection>
  </connection_0d>
</internal_catalog>
```

Homework: Consider what “misuse” or contractionary set of data can occur by this approach!

Homework: Create more examples, e.g. for gum drops, spot welds, rivets, <screw/> with attributes. → meeting directory in GitHub

Detailing XML Elements — Reference Syntax, Screw Example

The example suggests as a syntax for *referring to* an internal catalog item at `<connection_0d/>` level:

Default form — no change to / overwriting catalog data:

```
<connection_0d label="SCREW_100532" internal_catalog_item="screw_catalog_item_4711">
  <!-- Screw exactly as defined in internal catalog: -->
  <threaded_connection>
    <normal_direction x="2.9" y="0.1" z="0.0" />
  </threaded_connection>
  <loc> 1500.3809 838.75885 730.6529 </loc>
</connection_0d>
```

Minimal change — overwriting an attribute value:

```
  <!-- Screw exactly as defined in internal catalog, but with smaller torque: -->
  <threaded_connection internal_catalog_item="screw_catalog_item_4711" torque="70">
Put reference to conn_0d-level!
```

Detailing XML Elements — Reference Syntax, Screw Example

The example suggests as a syntax for *referring to* an internal catalog item at `<connection_0d/>` level:

Overwriting a child element:

```
<!-- Screw exactly as defined in internal catalog, but with larger washer: -->
```

```
<threaded_connection internal_catalog_item="screw_catalog_item_4711">
```

Put reference to conn_0d-level!

```
<normal_direction x="3.1" y="-0.1" z="0.0" />
```

```
<washer outer_diameter="30"/>
```

```
</threaded_connection>
```

Attention: Here, only the `outer_diameter` of the `<washer/>` is overwritten. Any other washer-parameter given from the catalog item stay effective.

Detailing XML Elements — Catalog Syntax, Bolt Example

The example extends to cases of *deeper structured* elements:

```
<internal_catalog>
  <connection_0d label="bolt_catalog_item_4712">
    <threaded_connection length="50" diameter="10"
      head_diameter="16" head_height="5" thread_length="35"
      torque="80" angle="30" pretension="1200" part_code="M10x50 12.9">
      <normal_direction x="0" y="0" z="-10"/>
      <washer outer_diameter="20" inner_diameter="10.3" thickness="1.5" attached="true"/>
      <bolt>
        <nut diameter="16." height="5" static_friction="0.8" clipped_to="4"/>
      </bolt>
    </threaded_connection>
  </connection_0d>
</internal_catalog>
```

Rename “internal_catalog” to “~~embedded_catalog~~” “connection_templates”
and attribute name “internal_catalog_item” to “template”!

Detailing XML Elements — Reference Syntax, Bolt Example

The internal catalog item can be *referred to* for example like this:

```
<connection_0d label="BOLT_100732" internal_catalog_item="bolt_catalog_item_4712">  
  <!-- Bolt exactly as defined in internal catalog, but with less torque, larger washer & nut: -->  
  <threaded_connection torque="70">  
    <normal_direction x="3.1" y="-0.1" z="0.0"/>  
    <washer outer_diameter="30"/>
```

Check, whether a washer may refer to a catalog item, too! → we need examples to better understand.

```
    <bolt>  
      <nut height="8"/>
```

Check, whether a nut may refer to a catalog item, too! → we need examples to better understand

```
    </bolt>  
  </threaded_connection>  
</connection_0d>
```

Rule (→ Word file): If the catalog item is a <screw/>, the referring element must not redefine it to a <bolt/> and vice versa!

To be extended to similar situations in case of future extensions of χMCF!

Contentual Action Items $\triangleq$ Next Steps:

▪ Standard document (Word file):

- Description of overall concept & objectives, intended application & motivation etc.
- Description of semantics for internal catalog
- Description of semantics for reference to a catalog item
- Definition of example use cases and in-document XML-examples

▪ XML & XSD — See discussion in [#105](#)!

- Sketching XML example file(s), incl. (w.i.p.)
 - precedence rule (w.i.p.) and dropped mandatory attribute,
 - internal (w.i.p.) and external file-based catalog
- Description of syntax for internal catalogs (w.i.p.)
- Description of syntax for **reference** to a catalog item:
 - internal (w.i.p.), external file-based, external database-based

- Drafting XML schema (XSD) and test cases / test example files

Homework: Collect use cases which can yield e.g. test cases, shared as Markdown files in GitHub at https://github.com/economidis-nick/createXSDforxMCF/tree/issues_61_94_95_105/VDA_FAT_AK_25/Meetings/2026-01-28_VideoConference

- Investigate xMCF elements and attributes for eligibility for template.
⇒ we need a rule, here! ⇒ Homework!



Next meetings

We need another focus group meeting!

- 2026-01-28, 15:00 CET — focus group (re-defined time slot)
- 2026-02-04, 15:00 CET — full group (new time slot)
 - objective: agree on XML format suggestions
- 2026-02-11, 15:00 CET — focus group on [#105](#) / [#61](#) (CF, NE, GH, SH, JB, ...) (re-defined time slot)
 - objective: create DOC suggestions & examples
- 2026-02-18, 15:00 CET — full group (new time slot)
 - objective: agree on DOC suggestions & examples

CF sent / will send out invitations.

